9.1 Sequences

A sequence is a list of numbers in a given order.

- Finite sequence: $\{a_n\}_{n=1}^{4} = \{1, 4, 7, 10\}$
- Infinite sequence: $\{a_n\}_{n=1}^{\infty} = \{3, 7, 11, 15, \ldots\}$

One can think of a sequence as a function restricted to positive integers.

$a(n) = a_n$

In our finite sequence above, we can say we have a sequence $\{a_n\}_{n=1}^{4}$ with $a_n = 1 + 3(n-1)$; and in the case of the infinite sequence above, we have a $\{a_n\}_{n=1}^{\infty}$ sequence with $a_n = 3 + 4(n-1)$.

E.g. $\{a_n\}$ with $a_n = \sqrt{n}$

$\{a_n\} = \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \ldots\}$

Representing Sequences

Definition: A **sequence** is a function a from the set of natural numbers to the set of real numbers. Instead of $a(n)$, we will usually denote the nth element in the sequence by a_n. We will use the notation $\{a_n\}$, or more precisely,

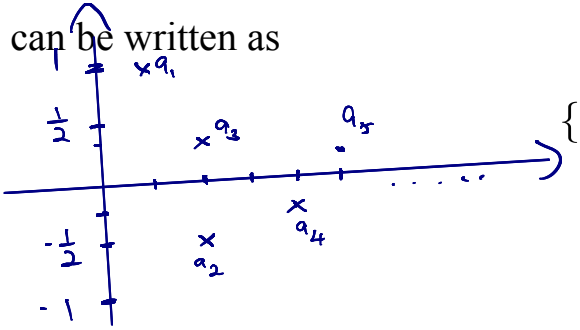
$$\{a_n\}_{n=1}^{\infty}$$

to denote a sequence.

As an example, observe that the sequence

$$\{a_n\} = \left\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5} \cdots\right\}$$

can be written as



$$\{a_n\} = \left\{\frac{(-1)^{n+1}}{n}\right\}.$$

Convergence and Divergence

Definitions The sequence $\{a_n\}$ **converges** to the number L if for every positive number ϵ there corresponds an integer N such that for all n,

$$n > N \quad \Rightarrow \quad |a_n - L| < \epsilon.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

The number L is said to be the limit of the sequence $\{a_n\}$, and we write $\lim_{n\to\infty} a_n = L$, or simply $a_n \to L$.

A sequence that converges is said to be **convergent**. Otherwise, the sequence is said to be **divergent**.

Example: The constant sequence 1, 1, 1, ... is convergent and the limit is 1.

For any given ε, we pick $N = 1$.

Example: The sequence $\{1/n\}$ converges to 0. That is, $\lim_{n\to\infty} \frac{1}{n} = 0$.

Given $\varepsilon > 0$, pick N such that $N > \frac{1}{\varepsilon}$,

then for any $n > N$, we have

$$|a_n - 0| = \left|\frac{1}{n} - 0\right| < \frac{1}{N} \text{ "} < \varepsilon \text{"}$$

Therefore, the seq. $\{a_n = \frac{1}{n}\}$ converges to 0.

Example:

The sequence $\{1, -1, 1, -1, ..., (-1)^{n+1}, ...\}$ is divergent.

We prove by contradiction. Let $a_n = (-1)^{n+1}$.

Suppose that $\{a_n\}$ converges to L.

Pick $\varepsilon = \frac{1}{2}$, then there exists N st for all $n \geq N$, we have

$$|a_n - L| < \frac{1}{2}$$

$$\Rightarrow L - \frac{1}{2} < a_n < L + \frac{1}{2}$$

For odd n, $a_n = 1$, so $1 < L + \frac{1}{2} \Rightarrow L > \frac{1}{2}$.

And so $a_n > L - \frac{1}{2} > 0$ for all $n \geq N$.

But this cannot be true for n even, because when n is even, we have $a_n = -1$, but $-1 \not> 0$. This is a contradiction, and hence $\{a_n\}$ is not convergent. $\therefore$ $\{a_n\}$ diverges.

Example: The sequence $\{\frac{n^2+1}{n^2+n}\}$ converges and

$$\lim_{n\to\infty} \frac{n^2+1}{n^2+n} = 1.$$

Given $\varepsilon > 0$, pick an N st $N > \frac{1}{\varepsilon} - 1$

Then for $n > N$,

$$|a_n - 1| = \left|\frac{n^2+1}{n^2+n} - 1\right|$$

$$= \left|\frac{1-n}{n^2+n}\right| = \frac{n-1}{n^2+n} < \frac{n}{n^2+n} = \frac{1}{n+1} < \frac{1}{N+1} \quad (\text{wat} < \varepsilon)$$

$$< \varepsilon$$

$\frac{1}{N+1} < \varepsilon$

$\Rightarrow N+1 > \frac{1}{\varepsilon}$

$N > \frac{1}{\varepsilon} - 1$

Therefore, $\{\frac{n^2+1}{n^2+n}\}$ converges to 1.

In some cases, a sequence diverges because it gets larger and larger. In particular, a_n will eventually become larger than any fixed number as n gets large. We use the following terms for such scenarios.

Definition The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all$n$ larger that N, we have $a_n > M$. If this condition holes, we write

$$\lim_{n\to\infty} a_n = \infty \text{ or } a_n \to \infty.$$

Similarly, if for every number m there is an integer N such that for all n larger that N, we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and write

$$\lim_{n\to\infty} a_n = -\infty \text{ or } a_n \to -\infty.$$

For example, the seq. $\{\sqrt{n}\}$ diverges to infinity.

and the seq. $\{-n^{3/2}\}$ diverges to $-\infty$.

Calculating Limits of Sequences

$a_n \to A$

$b_n \to B$

Theorem: Let $\{a_n\}$ and $\{b_n\}$ be be two convergent sequences of real numbers, and let A and B be the limits of $\{a_n\}$ and $\{b_n\}$respectively. Then the following rules hold.

$c_n = a_n + b_n$ $\quad c_n \to A + B$.

1. *Sum Rule*: $\lim_{n\to\infty} (a_n + b_n) = A + B$
2. *Difference Rule*: $\lim_{n\to\infty} (a_n - b_n) = A - B$ ($d_n = a_n - b_n$, then $d_n \to A - B$)
3. *Constant Multiple Rule*: $\lim_{n\to\infty} (k \cdot b_n) = k \cdot B$ (any k)
4. *Product Rule*: $\lim_{n\to\infty} (a_n \cdot b_n) = A \cdot B$
5. *Quotient Rule*: $\lim_{n\to\infty} \left(\frac{a_n}{b_n}\right) = \frac{A}{B}$ if $B \neq 0$

$a_n \to A \qquad b_n \to B$

Let $c_n = a_n + b_n$. Claim that $\underline{c_n \to A+B}$.

Given any $\varepsilon > 0$, we want to find N such that for all $n \geq N$, we have $|c_n - (A+B)| < \varepsilon$.

Fix an $\varepsilon > 0$.

Since $a_n \to A$, there is some N_1 st for all $n \geq N_1$, we have $|a_n - A| < \frac{\varepsilon}{2}$.

Similarly, there is some N_2 st for all $n \geq N_2$, we have $|b_n - B| < \frac{\varepsilon}{2}$.

Pick $N = \max\{N_1, N_2\}$. Then for $n \geq N$, we see that

$$|c_n - (A+B)| = |(a_n - A) + (b_n - B)| \leq |a_n - A| + |b_n - B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□.

Examples:

a. $\lim_{n\to\infty}\left(1-\frac{1}{n}\right) =$ $\left(\lim_{n\to\infty} 1\right) - \left(\lim_{n\to\infty}\frac{1}{n}\right) = 1 - 0 = 1$

b. $\lim_{n\to\infty}\left(\frac{5}{n^2}\right) =$ $\lim_{n\to\infty}(5)\cdot\lim_{n\to\infty}\left(\frac{1}{n}\right)\lim_{n\to\infty}\left(\frac{1}{n}\right) = 5\cdot 0\cdot 0 = 0$

c. $\lim_{n\to\infty}\left(\frac{4-7n^6}{n^6+3}\right) =$ $\lim_{n\to\infty}\frac{\left(\frac{4}{n^6}-7\right)}{\left(1+\frac{3}{n^6}\right)} = \frac{4\left(\lim_{n\to\infty}\frac{1}{n}\right)^6 - \lim_{n\to\infty}7}{\lim_{n\to\infty}(1) + 3\left(\lim_{n\to\infty}\left(\frac{1}{n}\right)\right)^3}$

Wrong $\overset{"}{=}$ $\frac{\lim(4-7n^6)}{\lim(n^6+3)} = -\frac{7}{1} = -7$

Remark:

1. One consequence of the above theorem is that if both the sequences $\{a_n\}$ and $\{b_n\}$ converge, their sum $\{a_n + b_n\}$ also converges. However, the converse is not true. For example,

$\{a_n\}$, $a_n = (-1)^n$ diverges

$\{b_n\}$, $b_n = (-1)^{n+1}$ diverges.

but $\{a_n + b_n\} = \{0\}$ converges to 0.

2. Another consequence is that every nonzero multiple of a divergent sequence $\{a_n\}$ is also a divergent sequence. Suppose that, to the the contrary, $\{ca_n\}$ converges for some $c \neq 0$. Then, write $k = \frac{1}{c}$, then by the theorem, the sequence $\{k \cdot ca_n\} = \{\frac{1}{c} \cdot ca_n\} = \{a_n\}$ converges, which is a contradiction.

($k \cdot ca_n$: convergent seq)

Therefore $\{ca_n\}$ diverges if $\{a_n\}$ diverges.

Theorem (The Sandwich Theorem for Sequences):

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences such that $a_n \le b_n \le c_n$ for all n beyond some index N. Suppose that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$ also.

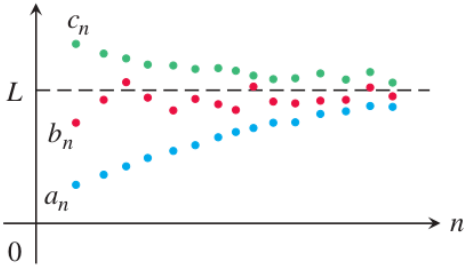


FIGURE 9.4 The terms of sequence $\{b_n\}$ are sandwiched between those of $\{a_n\}$ and $\{c_n\}$, forcing them to the same common limit L.

Example:

a. $\frac{\cos n}{n} \to 0$

$-\frac{1}{n} \le \frac{\cos n}{n} \le \frac{1}{n}$ for all $n \in \mathbb{N}$

b. $\frac{1}{2^n} \to 0$

$0 \le \frac{1}{2^n} \le \frac{1}{n}$

c. $\frac{1}{n\sqrt{n}} \to 0$

$0 \le \frac{1}{n\sqrt{n}} \le \frac{1}{n}$

Theorem (The Continuous Function Theorem for Sequences): Let $\{a_n\}$ be a sequence of real numbers. If $a_n \to L$ and if f is a function that is continuous at L and defined at all a_n, then $f(a_n) \to f(L)$.

Example: Show that $\sqrt{(n+1)/n} \to 1$.

We know that $\frac{n+1}{n} \to 1$, let $f(x) = \sqrt{x}$ then

$$f\left(\frac{n+1}{n}\right) \to f(1) = \sqrt{1} = 1.$$

Example: The sequence $\{2^{1/n}\}$ converges to 1.

We know $\frac{1}{n} \to 0$, let $f(x) = 2^x$

then $f\left(\frac{1}{n}\right) \to f(0) = 2^0 = 1$.

Using L'Hôpital's Rule

Theorem: Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x\to\infty} f(x) = L \quad \Rightarrow \quad \lim_{n\to\infty} a_n = L.$$

Example: $\lim_{n\to\infty} \frac{\ln n}{n} = 0$

Let $f(x) = \frac{\ln x}{x}$ Then $\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{x\to\infty} f(x)$

$= \lim_{x\to\infty} \frac{\ln x}{x}$ (of the form $\frac{\infty}{\infty}$)

$\overset{\text{l'Hôpital}}{=} \lim_{x\to\infty} \frac{1/x}{1}$

$= 0$

Example: Let $\{a_n\}$ be the sequence defined by $a_n = \left(\frac{n+1}{n-1}\right)^n$. Does the sequence converge?

Take natural log of a_n, $\ln a_n = \ln\left(\frac{n+1}{n-1}\right)^n$

$= n \ln\left(\frac{n+1}{n-1}\right)$.

$\lim_{n\to\infty} \ln a_n = \lim_{x\to\infty} x \ln\left(\frac{x+1}{x-1}\right)$ (by taking $f(x) = x \ln\left(\frac{x+1}{x-1}\right)$)

$\to \ln\left(1+\frac{2}{x-1}\right)$

$= \lim_{x\to\infty} \dfrac{\ln\left(\frac{x+1}{x-1}\right)}{1/x}$ (of the form of $\frac{0}{0}$)

$= \lim_{x\to\infty} \dfrac{\frac{x-1}{x+1}\cdot\left(-\frac{2}{(x-1)^2}\right)}{-1/x^2}$

$= \lim_{x\to\infty} \dfrac{2x^2}{x^2-1} = 2$.

apply l'Hopital rule again if you wish.

$\therefore$ The seq. $\{\ln a_n\}$ converges to 2.

$\Rightarrow$ The sequence $\{a_n\}$ converges to e^2.

[$a_n = e^{b_n}$, where $b_n = \ln a_n$.

Since $b_n \to 2$ $a_n \to e^2$].

Commonly Occurring Limits:

1. $\lim_{n\to\infty} \sqrt[n]{n} = 1$ $a_n = n^{1/n}$, $b_n = \ln a_n = \frac{1}{n}\ln n$

$\lim_{n\to\infty} b_n = \lim_{x\to\infty} \frac{\ln x}{x} = 0 \Rightarrow a_n \to e^0 = 1$

2. $\lim_{n\to\infty} k^{1/n} = 1$ for any fixed $k > 0$.

$\ln k^{1/n} = \frac{1}{n}\ln k \to 0$

3. $\lim_{n\to\infty} k^n = 0$ for any fixed k such that $|k| < 1$.

4. $\lim_{n\to\infty} (1 + \frac{k}{n})^n = e^k$ for any fixed k.

Take $\ln$. $\Rightarrow \lim n \ln(1+\frac{k}{n})$

5. $\lim_{n\to\infty} \frac{k^n}{n!} = 0$ for any fixed k.

$= \lim_{x\to\infty} x \ln(1+\frac{k}{x}) \to k$

$a_n = \frac{k^n}{n!}$ Claim: $a_n \to 0$.

Pick $M \in \mathbb{N}$ such that $M > k$, then $r := \frac{k}{M} < 1$.

For all $n > M$, observe that

$$a_n = \underbrace{\frac{k}{1} \cdot \frac{k}{2} \cdot \dots \cdot \frac{k}{M-1} \cdot \frac{k}{M}}_{\text{constant, bounded}} \cdot \frac{k}{M+1} \cdot \dots \cdot \frac{k}{n}$$

$$< C \cdot \underbrace{\frac{k}{M} \cdot \frac{k}{M} \cdots \frac{k}{M}}_{n-M \text{ times}}$$

$$= C \cdot r^{n-M}$$

$$= \underbrace{\frac{C}{r^M}}_{\text{constant}} \cdot \underbrace{r^n}_{\to 0 \text{ as } r<1} \longrightarrow 0.$$

Bounded Monotonic Sequences (brief)

Definition:

- A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n. ⇒ least u.b.
- A sequence $\{a_n\}$ is **bounded from below** if there exists a number m such that $a_n \geq m$ for all n. ⇒ greatest lower bound.
- If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is **bounded**. Otherwise, we say $\{a_n\}$ is an **unbounded** sequence.

Example: • $\{1, 2, 3, \cdots\}$ is bounded from below by any $M \leq 1$. but it is not bounded above

• $\{1, -1, 1, -1, 1, -1, \cdots\}$ is a bounded seq.

Definition: A sequence $\{a_n\}$ is **increasing** is $a_n \leq a_{n+1}$ for all n. The sequence $\{a_n\}$ is **decreasing** if $a_n \geq a_{n+1}$ for all n. We say a sequence $\{a_n\}$ is **monotonic** if it is either increasing or decreasing.

Eg.

- The seq. $\{\frac{1}{n}\}$ is decreasing.
- The seq. $\{k\}_{n\geq 1}$ is both increasing & decreasing (where k is a constant.)
- The seq $\{(-1)^n\}$ is not increasing & not decreasing.

Theorem (The Monotonic Sequence Theorem): If a sequence $\{a_n\}$ is bounded and monotonic, the sequence converges.

- If $\{a_n\}$ is bounded from above & increasing, then $\{a_n\}$ converges.
- If $\{a_n\}$ is bounded below & decreasing, then $\{a_n\}$ converges.

Example:

① The seq. $\{a_n = 1 - \frac{1}{n}\}$ is a bounded seq. since $0 \le a_n \le 1$ for all $n \ge 1$.

And $\{a_n\}$ is increasing. Hence, $\{a_n\}$ converges.

② $a_n = \frac{2^n 3^n}{n!}$: a bounded seq & decreasing.

$(a_n \ge 0)$

$$a_n = \frac{2^n 3^n}{n!} \ge \frac{2^{n+1} 3^{n+1}}{(n+1)!} \quad \text{for } n \ge 5.$$

$\therefore \{a_n\}$ converges.

Extra Examples:

1. $a_n = \frac{1-2n}{1+2n}$

2. $a_n = \frac{3^n 6^n}{2^{-n} n!}$

① Claim: $a_n \to -1$

Given $\varepsilon > 0$, pick N such that $\frac{2}{1+2N} < \varepsilon$. (OR just $N > \frac{1}{\varepsilon}$)

Then for all $n \geq N$, $|a_n - (-1)| = \left| \frac{1-2n}{1+2n} + 1 \right|$

$= \frac{2}{1+2n} \leq \frac{2}{1+2N} < \frac{1}{N} < \varepsilon$.

② $\lim_{n\to\infty} \frac{3^n \cdot 6^n}{2^{-n} n!} = \lim_{n\to\infty} \frac{(3 \cdot 2 \cdot 6)^n}{n!} = 0$

3. $a_n = \frac{(\ln n)^3}{n}$

4. $a_n = \sqrt{n}\sin(\frac{1}{\sqrt{n}})$

③ $\lim_{n\to\infty} \frac{(\ln n)^3}{n} = \lim_{x\to\infty} \frac{(\ln x)^3}{x}$

$\overset{\text{l'Hôpital}}{=} \lim_{x\to\infty} \frac{3(\ln x)^2}{x} \overset{\text{l'Hôpital}}{=} \lim_{x\to\infty} \frac{6\ln x}{x} = 0$.

④ $\lim_{n\to\infty} \sqrt{n}\,\sin\left(\frac{1}{\sqrt{n}}\right) = \lim_{x\to\infty} \sqrt{x}\,\sin\left(\frac{1}{\sqrt{x}}\right) = \lim_{x\to\infty} \frac{\sin\left(\frac{1}{\sqrt{x}}\right)}{\frac{1}{\sqrt{x}}} \quad \left(\frac{0}{0}\right)$

$\overset{\text{l'Hôpital}}{=} \lim_{x\to\infty} \cos\left(\frac{1}{\sqrt{x}}\right)$

$= 1$